NTNU
Kunnskap for en bedre verden

Department of Engineering Cybernetics

TTK4255 - Robotic Vision

Homework 1: Image Processing

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Feedback is appreciated on everything. You don't have to give detailed feedback on what is correct. We are curious if we have done something wrong or made it hard for ourselves.

Date

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1 Task 1

1.1 Task 1.1

Part 1:

1.1. a)

65	65	65	65		
65	65	65	65		
65	65	65	65	65	

$$* \begin{bmatrix} -0.5 & -1 & -0.5 \\ -1 & 7.0 & -1 \\ -0.5 & -1 & -0.5 \end{bmatrix} =$$

65	65	..	..
..	..	..	..
..	..		

It does nothing.

b)

1	1	1	1	1	1
1	1	1	1	1	1
1	1	1	1	1	1
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0

$$* \begin{bmatrix} -0.5 & -1 & -0.5 \\ -1 & 7.0 & -1 \\ -0.5 & -1 & -0.5 \end{bmatrix} =$$

1	1	1	1
3	3	3	3
-2	-2	-2	-2
0	0	0	0

It amplifies edges.

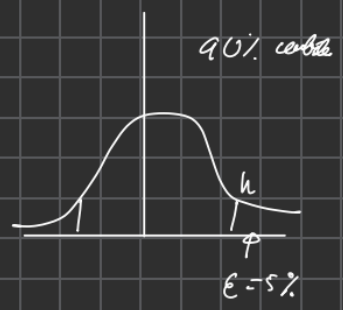
1.2 Task 1.2

$$g(x) < \varepsilon$$

$$g\left(\frac{h}{2}\right) \leq \varepsilon$$

1.2/

$$g(x) = \frac{1}{\sqrt{2\pi}\,\sigma} \cdot \exp\left(\frac{-x^2}{2\sigma^2}\right)$$



$$g(h) = \frac{1}{\sqrt{2\pi\sigma^2}} \cdot \exp\left(\frac{-h^2}{2\sigma^2}\right)$$

$$g\left(\frac{h}{2}\right) = \frac{1}{\sqrt{2\pi\sigma^2}} \cdot \exp\left(\frac{-\left(\frac{h}{2}\right)^2}{2\sigma^2}\right)$$

$$-\,\|\,- \cdot \exp\left(-\frac{1}{4} \cdot \frac{h^2}{2\sigma^2}\right) \leq \varepsilon$$

$$\exp\left(\cdot \frac{\alpha^2}{2\sigma^2}\right) \leq \varepsilon \cdot \sqrt{2\pi\sigma^2} \quad / \ln$$

$$-\frac{\alpha^2}{2\sigma^2} \leq \ln\left(\varepsilon\sqrt{2\pi\sigma^2}\right)$$

$$\frac{\alpha^2}{2\sigma^2} \geq -\ln\left(\varepsilon\sqrt{2\pi\sigma^2}\right)$$

$$\frac{\alpha^2}{2\sigma^2} \geq \ln\left(\frac{1}{\varepsilon\sqrt{2\pi\sigma^2}}\right)$$

$$\alpha \geq \sqrt{\ln\left(\frac{1}{\varepsilon\sqrt{2\pi\sigma^2}}\right) \cdot 2\sigma^2}$$

$$\frac{h}{2} \geq \sqrt{\ln\left(\frac{1}{\varepsilon\sqrt{2\pi\sigma^2}}\right) \cdot 2\sigma^2}$$

b) $\alpha \geq \sqrt{\ln\left(\frac{1}{\varepsilon\sqrt{2\pi\sigma^2}}\right)\cdot 2\sigma^2}$

$$\alpha \geq \sqrt{\ln\left(\frac{1}{\frac{1}{256}\sqrt{2\pi\, 3^2}}\right)\cdot 2\cdot 3^2}$$

$$\alpha \geq 8$$

$$\underline{K = 2k+1 = 17}$$

2 Task 2

2.1 Task 2.1

Width and height: (1280,720)

2.2 Task 2.2

i=1 is the green channel. This is the channel where the known green pixels are the brightest.

2.3 Task 2.3

Not easy using only thresholding. Seems like a value of 120 for the threshold gives a somewhat good distinction.

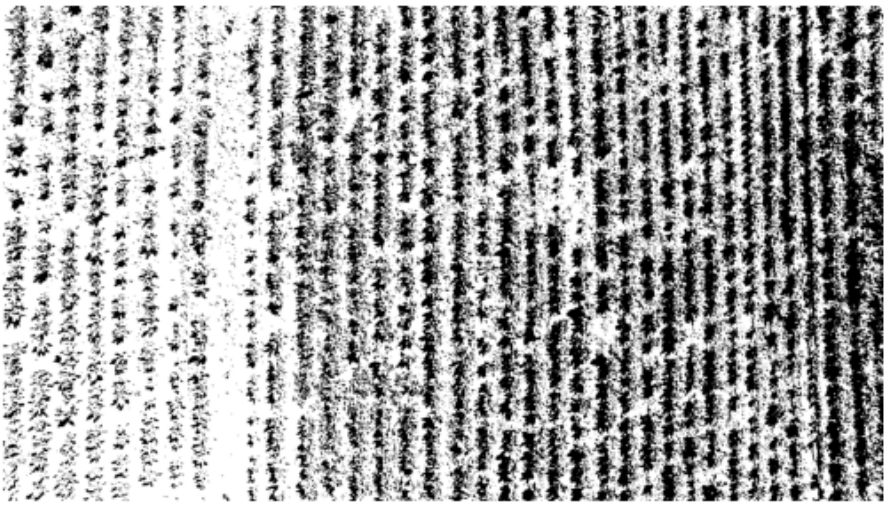


Figure 1: Task 2.3

2.4 Task 2.4

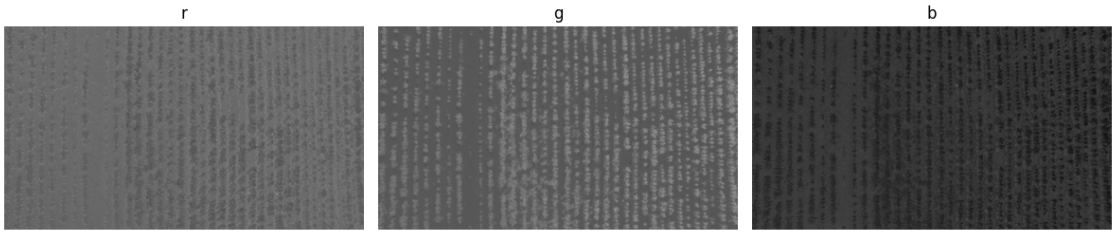


Figure 2: Task 2.4

2.5 Task 2.5

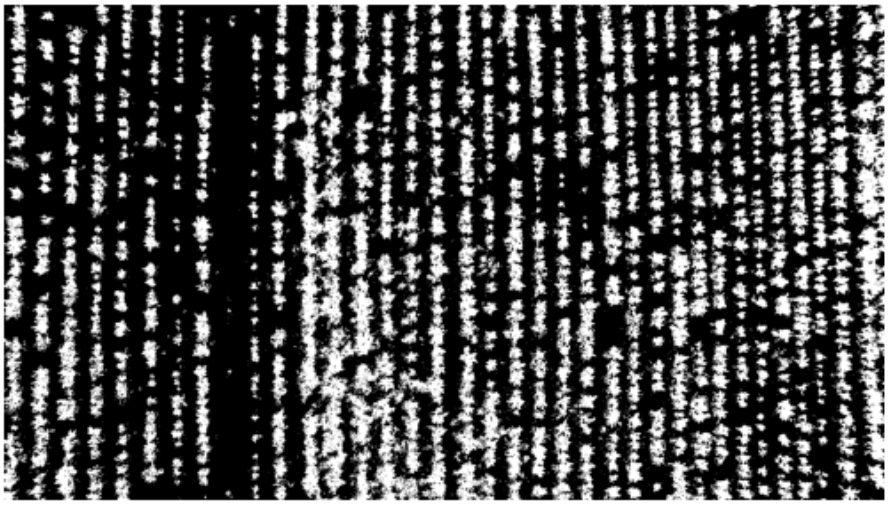


Figure 3: Task 2.5

3 Task 3

When increasing the blur value, the edges become more prevalent. It even detects edges that aren't always wanted.

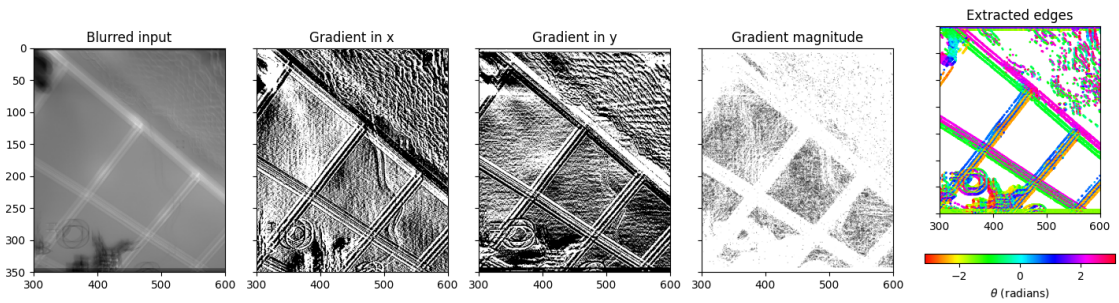


Figure 4: Task 3